

DAVID REY

Software Engineer | Backend Developer

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PROFILE

Software developer combining a professional game-development background with backend projects in Java, Spring Boot, Spring WebFlux, Node.js and NestJS. Builds REST APIs, authentication systems and relational/NoSQL persistence with clear architectural boundaries, automated tests and containerized delivery. Experienced in API integration, runtime troubleshooting and multidisciplinary collaboration.

TECHNICAL SKILLS

Backend: Java, Spring Boot, Spring WebFlux, Node.js, TypeScript, NestJS, REST APIs and reactive programming.

Architecture and security: hexagonal architecture, event-driven systems, separation of concerns, JWT authentication, access/refresh tokens and role-based authorization.

Data and quality: PostgreSQL, SQL, MongoDB, R2DBC, unit/integration/end-to-end testing, JUnit 5, Mockito and Swagger/OpenAPI.

Delivery: Docker, Docker Compose, GitHub Actions, GitHub Packages and Render. Integration: React, JavaScript, Unity and C#. Collaboration: Git, code reviews and Scrum.

SELECTED BACKEND PROJECTS

Game Transactions Backend | Node.js, NestJS, TypeScript, PostgreSQL

- Developed a REST backend for game purchases and inventory, with JWT authentication and separate authentication/payment providers supporting fake local-development flows and Steam-oriented integration.
- Implemented atomic, idempotent fulfilment to prevent duplicate item grants when purchase requests are retried. Added unit, integration and end-to-end tests, including container-backed database testing.
- Prepared Docker-based deployment and documented API contracts, configuration and the client/backend transaction flow. Kept client integration separate from backend implementation.

Web Hosting Management Platform | Java, Spring WebFlux, PostgreSQL, MongoDB

- Designed a reactive REST backend using hexagonal architecture, separating domain, application and infrastructure concerns for customer, form and project management.
- Implemented JWT access/refresh tokens, role-based authorization and authentication flows including login, token validation and password recovery.
- Connected a React frontend through REST APIs; used PostgreSQL for user data and MongoDB for forms and projects. Containerized the backend and databases with Docker Compose.

Blackjack API | Java, Spring WebFlux, R2DBC, Docker

- Built a reactive Blackjack backend with PostgreSQL and MongoDB persistence and Swagger/OpenAPI documentation.
- Automated application builds, Docker image publication and deployment to Render through GitHub Actions; configured environment-based database connectivity.

PROFESSIONAL EXPERIENCE

Eongate | Technical Artist and VR Developer

April 2026 - July 2026 | Remote

- Independently researched and resolved real-time performance problems for an Unreal Engine VR project, improving scene performance from approximately 2 FPS to a stable 120 FPS.
- Redesigned the rendering approach and worked directly with artists to establish performance-aware production workflows.

Arcadio Lab | Unity Programmer and Technical Artist

July 2024 - September 2025 | Barcelona

- Developed and maintained C# gameplay systems in multidisciplinary Scrum teams. Diagnosed software, API and runtime issues across production builds.
- Refactored external API data flows to eliminate unnecessary requests and reduce token consumption, avoiding the need for a higher service plan.
- Participated in pull requests and code reviews; collaborated with artists and designers to troubleshoot performance issues and improve technical workflows.

Forged Horizon | Unreal Engine VR Developer and Technical Artist

September 2023 - June 2024 | Barcelona

- Implemented interaction and gameplay systems with Blueprints in a five-person team. Golden Hand won Best Project of the Year at Deià in 2024.

Grupo 3 Studio | Unity Programmer and Technical Artist

September 2022 - July 2023 | Barcelona

- Sole programmer in a six-person team. Separated content from gameplay logic using ScriptableObjects so designers could configure missions and dialogue without programming.
- Built state-machine-based player and enemy behavior and profiled gameplay systems to improve runtime stability.

ADDITIONAL INTEGRATION PROJECT

Conversational AI NPC | Unity, C#, REST APIs

- Built a VR prototype connecting speech recognition, language-model services, speech synthesis and lip synchronization. Used decoupled, event-driven services with interchangeable local/API implementations.
- Implemented a state machine to prevent invalid concurrent actions and separated character configuration from conversation logic.

EDUCATION

IT Academy - Barcelona Activa

Specialization Course in Java Spring Boot Development | September 2025 - March 2026

Deià Escola d'Art i Disseny

Audiovisual Graphics - Virtual Reality and Gamification | 2023 - 2024

Audiovisual Graphics - Video Games and Virtual Environments | 2020 - 2023

LANGUAGES

Spanish: Native | Catalan: Native | English: Advanced